



Research Paper

Thermal interference in aquifer thermal energy storage: Insights from stochastic surrogate modelling of well placement

Elena Petrova ^a, Mauro Cacace ^a, Stefan Kranz ^a, Arno Zang ^a, Guido Blöcher ^{a,b}

^a GFZ German Research Center for Geosciences, Potsdam, Germany

^b Technical University Berlin, Department of Engineering Geology, Berlin, Germany

ARTICLE INFO

Keywords:

Aquifer Thermal Energy Storage (ATES)
Well placement
Stochastic modelling
ATES in urban areas
Optimal use of subsurface
Heat recovery factor

ABSTRACT

Hot Temperature Aquifer Thermal Energy Storage (HT-ATES) systems have gained interest due to their high storage capacity, cost-effectiveness, and environmental sustainability. Especially in urban areas with surplus heat, geothermal district heating becomes highly beneficial. Wells in ATES must be spaced to avoid thermal interference, but urban space and regulations often constrain well placement. In this numerical study, we provide placement criteria for well doublets in urban areas. In contrast to previous model approaches, the approach optimizes the trade-off between thermal interference and well spacing by maximizing energy recovery and minimizing well distance. To minimize computational costs, we designed a workflow where the full order physics-based model was replaced by a Random Forest Classifier (RFC) that predicts thermal interference with 97% accuracy accounting for technical and geological uncertainties in a typical deep HT-ATES system. The model was applied to an ATES system planned in Berlin-Adlershof. Results show that the well distance between 130 and 160 meters is optimal from the thermal and hydraulic perspective over a wide range of parameters. The global sensitivity analysis of the surrogate model reveals that influential parameters for thermal interference are well placement, aquifer thickness, flowrates, charging temperature, permeability, and porosity. By providing insights into key design priorities and enabling an explicit understanding of thermal interference, the developed RFC is an efficient tool for pre-evaluating possible ATES configurations and supporting urban planners' decision making.

1. Introduction

Decarbonization and elimination of greenhouse gas emissions have been in focus of the United Nations society since the last 20 years [1]. As one of the solutions, HT-ATES systems are an effective and relatively low-cost technology for reusing and storing energy [2]. HT-ATES enables high-efficiency, seasonal thermal energy storage while offering low operational cost, minimal land use, and near-zero emissions, which makes it a scalable and cost-effective solution for integrating into district heating and cooling systems.

In high temperature ATES systems, excessive heated water is injected into the subsurface in the summer time while the colder water is extracted for cooling the system. During winter, the process is reversed: the previously injected hot water is extracted from production wells and circulated through heat exchangers to heat buildings, with the cold water being reinjected into the aquifer. This ATES technology has been successfully implemented in over 1000 sites worldwide, mainly in the Netherlands [3], delivering an attractive alternative to carbon energy sources. Especially in urban areas which are characterized by

excessive heat sources, ATES has the potential to play a decisive role in the energy storage [see e.g. 4,5].

While the appeal of energy storage with ATES is significant, its successful implementation requires a thorough understanding of subsurface properties, alongside a careful planning and management of technical solutions [1,6]. Several analytical solutions and key performance characteristics have been developed to evaluate the efficiency of an ATES project [7,8], among those the most widely used is the Heat Recovery Factor (HRF) [9–11]. Physics-based numerical models are typically used for simulation of complex natural conditions such as aquifer heterogeneity and reactive behaviour [12–14], prediction of ATES efficiency under various ambient flow conditions [15] both at the district scale [16] and city scale [17], as well as targeting interacting systems [18]. The typical outcome of such models are temperature and pressure evolution curves over the whole lifetime of ATES system, which allows a proper quantification of the extracted energy and aquifer performance.

* Corresponding author.

E-mail address: elenape@gfz.de (E. Petrova).

<https://doi.org/10.1016/j.applthermaleng.2025.127012>

Received 20 March 2025; Received in revised form 17 May 2025; Accepted 29 May 2025

Available online 13 June 2025

1359-4311/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

From a design perspective, natural parameters such as aquifer properties (porosity, permeability, aquifer thickness, thermal conductivity, and heat capacity) and relatively low ambient flow velocity are necessary for high ATES efficiency [5,19–21]. At the same time, technical parameters such as distance between wells, charging temperature and flowrate, and duration of injection and extraction cycles define the boundary conditions to be operated in the injection and extraction wells [22–24].

To ensure optimal performance and efficiency, both natural and technical parameters of ATES systems have to be carefully estimated [25]. Sommer et al. [26] highlighted the importance of an optimal well spacing. Recent modelling studies [27,28] demonstrated the relevance of the optimization of the initial ATES design, which entails an accurate characterization of the system's parameters. Besides technical design parameters, natural aquifer parameters, are influenced by various uncertainties due to limited subsurface data [29,30].

The ensemble-based stochastic modelling has shown its capability in uncertainty reduction and decision making [29,31]. The challenge of stochastic approaches is the high computational cost of a single simulation. To overcome the bottleneck of the high computational cost of stochastic simulations on a spatially resolved hydraulic–thermal model, surrogate modelling approaches have been developed [32,33]. Regression-based surrogate models were used to target different questions of optimal design [8,34–37] and sensitivity analysis [38]. Rohmer et al. [19] proposed using multifidelity cokriging approach for fast yet accurate ATES characterization for technical ATES parameters and aquifer properties to predict the evolution of well temperatures for different aquifer properties. By using a surrogate modelling approach [28] were able to determine optimal production rate for a site in crystalline rocks.

Despite significant advancements in uncertainty quantification and regression analysis for optimizing ATES design, only limited amount of studies focused on the optimal well spacing by accounting for both technical and natural uncertainties [8,39,40]. Current design approaches often rely on generalized analytical equations such as multiplied thermal radius [26,27] and only define the minimum distance between the wells while ignoring the maximum distance estimation.

This study aims to bridge this gap by newly introducing both hydraulic and thermal criteria on the well placement in the HT-ATES doublet. While the thermal criteria define the minimum distance between the wells, the hydraulic criteria are capable to determine the maximum distance, thus leading to the range of possible distances for the doublet system. Based on an ensemble of spatially resolved numerical models, we confirmed the hypothesis of asymptotic development of HRF over the broad parameter ranges, and suggested a possible analytical form of it. To speed up the model evaluation, we developed a model reduction technique that retains the complexity of full-scale models while facilitating efficient predictions. We used the key characteristic of the reduced physics-based model to create a computationally effective surrogate model. To predict ATES efficiency and optimize the well distance, a Random Forest Classifier [40,41] was trained on an ensemble of 1500 spatially resolved numerical models. The methodology was applied to an ATES project in Berlin, Germany, where spatial constraints and regulatory requirements make optimal well placement crucial for system success. The outcome was compared with the conventionally used thermal radius approach for the well distance selection. The results offer a scalable solution for ATES design in similar urban contexts globally.

2. Methods

The method section includes the description of the site (Section 2.1), the physics-based ATES model developed in MOOSE-GOLEM Framework (Section 2.2), the key performance characteristics which lead to reduction of the model order to address model interference (Section 2.3) and the model reduction (Section 2.4).

2.1. Study area

The ATES system considered in this study is analogous to the planned ATES in the southeast of Berlin district of Adlershof. This ATES utilizes fine-grained Jurassic sandstones to store and provide heat for the district heating network. The Hettang aquifer sediments are predominantly weakly consolidated, fine-grained sandstones with a high quartz content and varying organic fractions (coal flakes) [16]. The planned ATES will consist of two wells with filter zone penetrating the full aquifer thickness at the depth intervals 360 to 390 m (Fig. 1).

The ATES system in Berlin Adlershof is targeting an aquifer which has a thickness of 30 to 60 m and consists of transgression–regression cycles of clay, siltstones, and sandstones. The Hettang Jurassic aquifer is isolated from the above positioned artesian aquifer by the Rupelian clay layer with thickness variation from 40 to 120 m and the screened thickness at the Adlershof site is more than 100 m [42]. With the ambient temperature of 24 °C and high salinity, the targeted for HT-ATES Hettang aquifer is well isolated from the artesian aquifer [43]. The Hettang aquifer is characterized by slow (less than 0.1 m/year) ambient groundwater velocity, which makes it plausible for geothermal storage utilization [44].

Preliminary site investigations, complemented by regional data, indicated that the permeability ranges from $1 \cdot 10^{-13}$ to $1 \cdot 10^{-12}$ m², with porosity between 0.1 and 0.4 [16]. For modelling purposes, a permeability range of $1 \cdot 10^{-14}$ to $5 \cdot 10^{-12}$ m² is considered to encompass most realistic porous aquifers suitable for ATES systems. To ensure a comprehensive approach, no empirical relationship between porosity and permeability was imposed, allowing these parameters to be treated independently. The initial assessment suggests that the Hettang aquifer at Adlershof is approximately 60 m thick, although it remains uncertain pending further drilling results. To account for lateral variations in aquifer thickness and associated uncertainties, a wide range of thicknesses from 10 to 100 m were considered in the study. The parameter ranges were extended to account for parametric uncertainty (Table 1).

The urban setting of the ATES site presents specific challenges. Due to the dense urbanization and limited access to the properties, the choice of well distance in Adlershof is limited. On the existing drilling site, the maximum distance possible in the reservoir depth is approximately 100 metres. Because of the complex ownership and leasing conditions in the densely populated area, larger distances require additional approvals. Therefore, careful planning of the well placement is crucial to optimize the system's efficiency while complying with regulatory requirements. From the technological perspective, the Adlershof site offers a range of possible technical solutions for the ATES system, as summarized in Table 1. The considered charging temperatures vary between 70 to 100 °C, and charging/extraction rates of 50 to 150 m³/h.

In such HT-ATES systems, optimizing the distance between wells is crucial for maximizing energy performance. Wells placed too close reduce pressure drop and improve hydraulic efficiency, but risk thermal interference, which lowers energy recovery. Conversely, spacing wells too far apart minimizes thermal interference but increases flow resistance and drawdown, potentially violating hydraulic constraints. Thus, both minimum and maximum well distances must be considered to balance thermal and hydraulic performance.

2.2. Physics-based ATES simulation

The ATES is simulated by system of PDE describing flow and heat transport in a fully saturated aquifer. The flow of fluid through a porous medium is governed by Darcy's law:

$$\mathbf{q} = -\frac{\mathbf{k}}{\mu} (\nabla p - \rho_f \mathbf{g}), \quad (1)$$

where $\mathbf{q}$ is the Darcy velocity (m/s), $\mathbf{k}$ is the permeability tensor (m²), μ is the fluid dynamic viscosity (Pa s), p is the fluid pressure (Pa), ρ_f is the fluid density (kg/m³), $\mathbf{g}$ is the gravitational acceleration (m/s²).

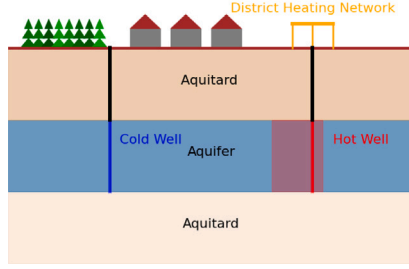


Fig. 1. Conceptual model of ATEs.

Table 1

Ranges of technical and natural ATEs parameters used for stochastic modelling.

Parameters	Unit	Min	Max
Technical parameters			
Well distance	m	50	400
Charging temperature	°C	60	100
Flowrates	m ³ /h	50	150
Natural parameters			
Permeability	m ²	1 · 10 ⁻¹⁴	5 · 10 ⁻¹²
Porosity	–	0.1	0.4

The mass conservation equation for fluid flow in the porous medium is given by:

$$\frac{\partial (\phi \rho_f)}{\partial t} + \nabla \cdot (\rho_f \mathbf{q}) = s, \quad (2)$$

where ϕ is the porosity of the medium (-), s denotes sources or sinks, and $\frac{\partial(\cdot)}{\partial t}$ is the time derivative.

The heat transport equation, considering both advection and conduction, is:

$$\frac{\partial (\rho_b c_p T)}{\partial t} + \nabla \cdot (\rho_f c_{pf} \mathbf{q} T - \lambda \nabla T) = f, \quad (3)$$

where T is the temperature (K), ρ_b is the bulk density (kg/m³), c_p is the specific heat (J/(kg K)), where the subscripts f and b stand for fluid and bulk, respectively, λ is the thermal conductivity tensor, which for porous media transport equals $\lambda = \phi \rho_f c_{pf} \mathbf{D}_T$ with $\mathbf{D}_T$ being the thermal dispersion tensor (m²/s), and f is denoting source/sink terms.

The initial conditions refer to the natural temperature gradient in the 3D domain:

$$T(\mathbf{x}, \mathbf{y}, \mathbf{z}, t = 0) = T_{\text{nat}}(\mathbf{x}, \mathbf{y}, \mathbf{z}), \quad (4)$$

To condition the model to the natural state, Dirichlet boundary conditions were applied at the lateral and bottom boundaries to represent undisturbed geothermal conditions [45,46], while no-flow (Neumann) conditions are assumed at the top boundary to reflect low permeability of an aquitard. Internal boundary conditions for ATEs are given by 0.5 years injection and extraction cycles in warm and cold wells. The modelled system will focus on a basic ATEs setup consisting of two wells penetrating the full aquifer thickness. The excessive hot water is injected into warm well to store the heat while at the same time, aquifer groundwater is extracted from the cold well. After half a year, the previously injected warm water is extracted from the warm well to provide heating, and the cooled (until aquifer temperature) water is injected into the cold well. Together, injection and extraction periods from a cycle which is repeated (for example, seasonally) during ATEs functioning (Fig. 1).

The study focused on ATEs where ambient groundwater flow is negligible. Lateral boundary conditions were based on global modelling results, considering both expected and observed data from nearby locations. No-flow boundary conditions were applied along the bottom



and top horizontal boundaries to simulate impermeable rock layers. The aquifer was assumed to have homogeneous isotropic properties within the model domain.

The computational domain was discretized using a mesh of vertically extruded triangular prisms, with horizontal spacing varying from 0.1 m around the wells to 50 m at the model's boundary to ensure the high resolution around the well. This non-uniform grid was generated using the open-source software GMSH [47] (Fig. 2, left). Coupled flow and heat transport equations within porous media were solved using the finite element software MOOSE-GOLEM [48, 49]. The aquifer was simulated as a single material layer with homogeneous parameters (porosity, permeability, heat capacity). This simplification is commonly used to reduce model complexity and computational load, especially at ATEs planning stages when site-specific heterogeneity data is limited due to limited amount of deep boreholes reflecting on the spatial permeability–porosity distribution. Previous studies e.g., Frick et al. [46] show that homogeneous models effectively capture dominant transport behaviour. While small-scale features like layering or anisotropy are simplified, the approach offers a solid basis for system understanding and can be extended with spatial variability in future work. Density and viscosity were set temperature and pressure dependent updated according to IAPWS-97 [50]. For compliance, we neglected interactions with adjacent low hydraulic conductivity layers. Additionally, we did not consider storage periods where injection and extraction phases are separated by intervals of inactivity.

During the simulation, the hot water is injected at a constant flowrate into the aquifer while the ambient temperature water is produced from another well, and after 6 month the flow direction in the wells is reversed. The simulation included ten yearly cycles of continuous injection and extraction, each phase lasting six months. This 10-year period was chosen based on convergence studies, which indicate that system behaviour stabilizes over this time window [51]. The performed simulations were stabilizing within the 6 to 10 years period. We made use of an adaptive time stepping based on the computed Courant number for the system, with a default time step size of 1 day. Charging and extraction flowrates were kept constant throughout the simulation. Key model parameters, including porosity, permeability, charging temperatures, flowrates, aquifer thickness, and well spacing are detailed further in Section 5. The model outputs included a fully 3D, spatio-temporally resolved field of temperature and pressure distributions (see Fig. 2, right).

2.3. Key performance characteristics

One of the most broadly used characteristics of ATEs efficiency is the Heat Recovery Factor (HRF) which is computed as the ratio between the produced and injected energy [9,52]:

$$HRF = \frac{\int_{t=1}^{t_2} \dot{V}_{\text{out}}(t) \rho c_p (T_{\text{out,w}}(t) - T_{\text{out,c}}(t)) dt}{\int_{t=0}^{t_1} \dot{V}_1(t) \rho c_p (T_{\text{in,w}}(t) - T_{\text{in,c}}(t)) dt}, \quad (5)$$

where $\dot{V}$, ρ , c_p , T , t are flowrate (m³/h), density of groundwater (kg/m³), specific heat capacity of groundwater (kJ/kgK), temperature

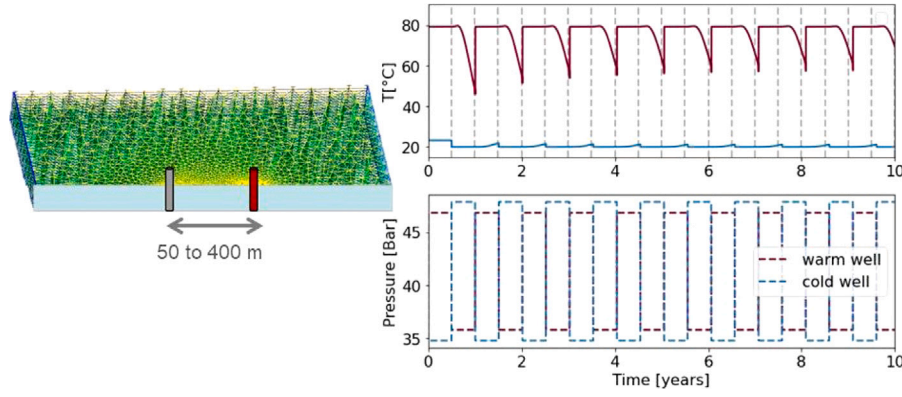


Fig. 2. Model setup: mesh of extruded triangular prisms with a vertical exaggeration of 10 (left); model output: timeseries of pressure and temperature in the cold and warm well (right).

(° C), and time (0 corresponds to the beginning of injection of the hot water, 1 is stop of injection and start of the extraction, 2 is the end of the cycle). The subscripts out and in indicate discharging energy and charging energy and w, c stand for warm well and cold well respectively.

An HRF value of 1 indicates that all injected energy can be recovered from the aquifer. An HRF value lower than one indicates that part of the energy is not recovered, due to for example, dispersive–diffusive losses, conditioning of the aquifer due to heating the solid phase of the aquifer, or thermal interference. An ATES system must be designed such that the loss term is minimized while HRF is maximized.

Another important metric for evaluating ATES system performance is the Stored Energy (SE). SE quantifies thermal losses due to interference with the cold well, defined as the ratio between the energy injected into the aquifer (represented by the rectangular area in Fig. 3, top-left) and the energy stored, which is calculated as the difference between the injected temperature and the temperature in the cold well during the charging phase (Fig. 3 top-left and bottom-left). For equal injected and extracted volumes SE can be expressed as:

$$SE = \frac{\int_{t=t_0}^{t_1} \rho c_p (T_{w,in} - T_{c,out}) dt}{\int_{t=t_0}^{t_1} \rho c_p T_{w,in} dt} \quad (6)$$

A SE value of 1 indicates that all injected energy is successfully stored, while SE values below 1 reflect energy losses due to thermal interference. When combined with HRF estimates, these two parameters are crucial for assessing the overall technical performance of ATES systems.

The typical characteristic parameter to limit the thermal influence in a ATES system is the thermal radius [9,24]:

$$R_{th} = \sqrt{\frac{c_w \dot{V} t}{c_a \pi H n}} \quad (7)$$

where $\dot{V}$, t , c_w , c_a , H , and n are charging flowrate, duration of the injection, heat capacity per unit of water, bulk heat capacity per unit volume (solid plus fluid), aquifer thickness, and porosity correspondingly.

This method assumes that the thermal front in the aquifer has a cylindrical shape. The engineering approach is that thermal interference can be effectively avoided if the distance between the wells exceeds 1.5 to 3 times the thermal radius [1].

Productivity Index (PI) characterizes well performance of ATES and shows how much of pressure change is induced by the current extraction flowrate:

$$PI = \frac{\dot{V}}{4p} \quad (8)$$

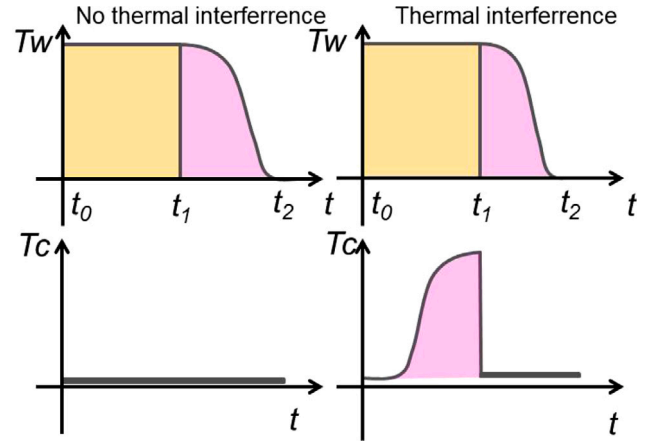


Fig. 3. A scheme of the temperature evolution in cold and warm wells during a single injection–extraction cycle: (left) without thermal interference, where heat interference is minimized, demonstrating optimal temperature retention in the wells; (right) with thermal interference, showing heat transfer between wells, resulting in diminished thermal storage efficiency; subscripts w and c correspond to warm well and cold well, respectively.

The drawdown in the ATES well doublet is the super positioning effect of the cold and warm well. For an isotropic homogeneous aquifer is described by the superposition of Theis functions:

$$\Delta p = \frac{\rho_f g \dot{V}}{4\pi T} (W(u_1) - W(u_2)) \quad (9)$$

where T is the transmissivity of the aquifer (m^2/s), $W(u)$ is the Theis well function, $u = \frac{r^2 S}{4Tt}$ is the dimensionless time parameter, r is the radial distance from the pumping well (m) with well diameter for the extraction well and distance between the wells for the injection well, S is the storage coefficient (-). The injection well increases pressure, while the extraction well decreases it, with their interaction influenced by well placement, flowrates, and aquifer properties.

2.4. Model order reduction

To quantify the technical feasibility of an ATES system, we would need to compute for each model realization time histories of HRF and SE indices (Fig. 4 bottom) based on computed charging/discharging temperature timeseries (Fig. 4 top). To assess the technical feasibility of ATES, it is necessary to compute timeseries HRF and SE for each model realization. These indices are typically derived from time series data representing the temperature variations during injection and production. Training a surrogate model to predict these indices directly from

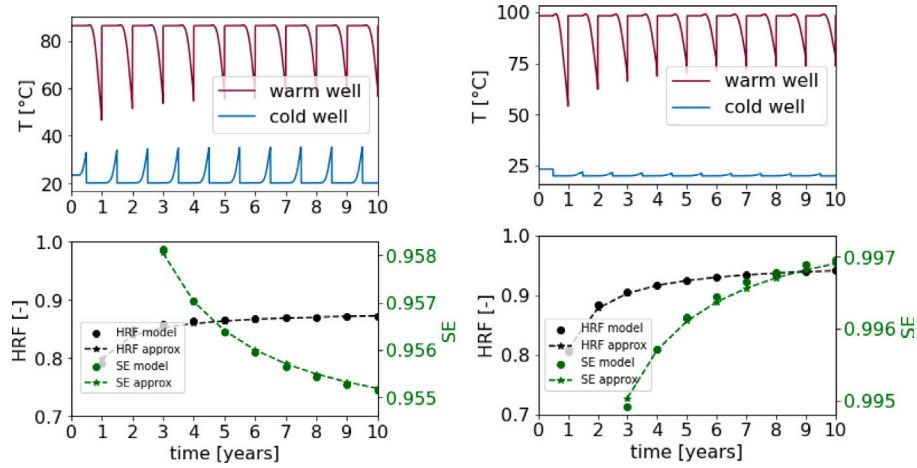


Fig. 4. Left: Model output of a 3D physics-based model with distance between the wells 59 m, charging temperature 86 °C, charging flowrate 103.3 m³/h, aquifer permeability 5e−14 m², porosity 0.3, thickness 44.7 m. Right: Model output of a 3D physics-based model with distance between the wells 120 m, charging temperature 98 °C, charging flowrate 95.4 m³/h, aquifer permeability 7e−14 m², porosity 0.27, thickness 28.6 m. For both cases, the Top plot refers to the temperature time series in the boreholes, the bottom plot shows HRF and SE development over the charging–extraction cycles analytical fit of the HRF and SE evolution: filled circles refer to the values obtained from the model, dashed lines refer to the analytical fit (Eq. (10)–(11)).

time series data presents challenges. While multiple-output surrogate models are increasingly common, studies have shown that training complexity and data requirements grow with the increasing dimensionality of the output space, which significantly impacts computational effort and model accuracy [53,54].

To enhance the computational efficiency, it is advantageous to reduce the model order rather than using time series data directly [55]. Previous studies found out that the efficiency and HRF have asymptotic behaviour [51] (Fig. 4, dashed lines at the bottom plot). In this work, we propose to fit the HRF and SE time series with simple asymptotic equations:

$$HRF(t) = \frac{a \cdot t}{b + t}, \quad (10)$$

$$SE(t) = \frac{c \cdot t}{d + t}, \quad (11)$$

where t refers to time since ATEs start, a , b , c , and d are empirical coefficients.

By making use of those analytical coefficients, the data complexity is substantially reduced. In such a way, each model produces only two scalars which allow for a full reconstruction of the HRF or SE time series. Fig. 4 - top presents an example of a single model output with temperature evolution in the warm and cold well. The corresponding HRF and SE computed for each cycle from Eq. (5)–(6) are shown as solid circles, while the approximated values, obtained by fitting a and b (c and d for SE) coefficients to Eqs. (10)–(11), as dashed curves. We note a perfect fit of the derived asymptotic functions (Eq. (10) and (11)) to the model output. Fitting results for all models are discussed in Section 4.1.

To validate Eq. (10)–(11) across a broad parametric space, the fit of the asymptotic approximation was compared with an ensemble of 1500 full model runs with randomly sampled parameter combinations. To fit the function to the model output, we used non-linear least squares method utilizing Levenberg–Marquardt algorithm implemented in “curve_fit” function in Python [56]. In such a way, we reduced the fully resolved model to a simple asymptotic approximation (Eq. (10)–(11)) without information loss.

The time derivative of SE which is represented by d coefficient serves as a crucial indicator in the context of thermal interference prediction. A negative derivative suggests a decrease in the system efficiency due to thermal interference (Fig. 4, bottom-left).

Consequently, optimal well placement focuses on monitoring this derivative as a key output metric in model simulations. To target the

optimal well spacing, the model can be thus further reduced into a binary output where 1 ($d < 0$) corresponds to the case when thermal interference is causing a decrease in efficiency and drop in SE (Fig. 4, bottom-left, and 0 ($d \geq 0$) the opposite, when no decrease in HRF and efficiency is observed (Fig. 4, bottom-right).

3. Surrogate model setup and optimization

In Section 3.1, we present the workflow for ensemble generation using quasi-random sampling, with adjustments for key parameters. In Section 3.2, we detail the approach of training a Random Forest Classifier (RFC) and compute the probability of thermal interference, and outline the global sensitivity analysis technique. In Section 3.3, we describe the infill procedure and a confusion matrix as a quality assessment of the surrogate model.

3.1. Ensemble generation

The surrogate model has to cover all parametric space and reflect on the parameter connections of the physics-based model, which requires high prediction accuracy. We enable this by creating a relatively large ensemble of parameter combinations and sampling uniformly the parametric space. This is done by using a quasi-random Halton sequence with a uniform prior probability density function. Physical parameter values of the model range over different scales and have different units (Table 1).

The permeability and well distance values were log-transformed to smooth over the parameter space. In case of the well distance which varies from 50 to 400 m, we gain more samples from the small distance values, which tend to produce thermal interference. This way, the sampling resolution around small well distances was increased. Then, parameters were normalized in their ranges in order to homogenize each of them between 0 and 1. The model is constrained by the appearance of turbulent flow in the vicinity to the well. This was avoided by limiting the charging and extraction rates based on the Reynolds number.

3.2. Calibration of a surrogate model

A single run of a spatially resolved numerical model requires several hours on a standard PC with 8 cores at 2.4 GHz, 16 GB RAM, Ubuntu 20.04. Considering the high computational cost of a single model, this

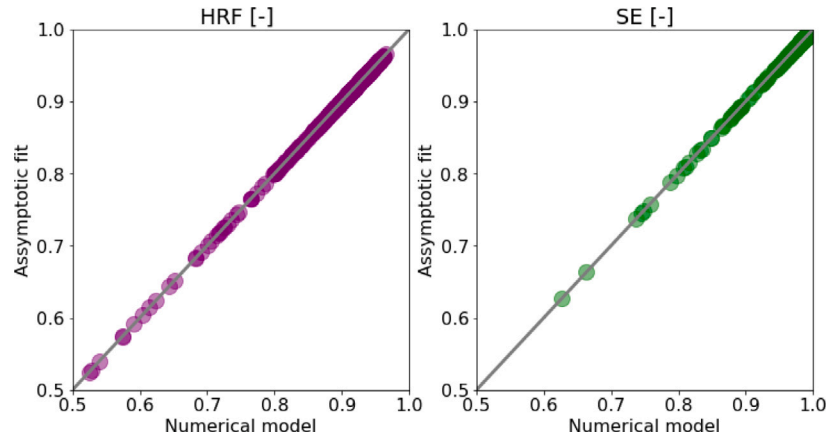


Fig. 5. Parity plot for model reduction: mean HRF (left) and SE(right) values from the 3D MOOSE-GOLEM simulation (abscissa) against corresponding mean values acquired from the asymptotic approximation (ordinata). The solid black line corresponds to 1:1 ratio.

puts the global sensitivity of HT ATEs simulation under constraint of computational resources.

To address this issue, using the proxy models, also referred to as surrogate models, has been shown to be advantageous for maintaining model accuracy while reducing computational efforts [19,28,35].

In this study, we employed RFC to model a binary classification problem with six input parameters: permeability, porosity, aquifer thickness, charging temperature, flowrate, and distance between two wells. The algorithm [41] is an ensemble learning technique that constructs a multitude of decision trees during training and outputs the mode of the classes for classification tasks or the mean prediction for regression tasks. The RFC offers several advantages for our purpose in comparison to other ML methods. As such, RFC is faster to train than Gradient Boosting methods since it trains trees in parallel which speeds up the training, and has more intuitive hyperparameter tuning, yet allowing the sensitivity analysis [57,58]. The RFC handles larger datasets better than Support Vector Machines and does not necessarily require normalization, and finally, it is targeting to use categorical data which is the goal of the study [58–60]. This method is particularly effective in handling high-dimensional data and mitigates the risk of over-fitting.

The model input parameters were used as input features for the RFC with a binary output. We configured the RFC with 100 trees, where each tree was built using a subset of the data and features, selected through bootstrapping and random feature selection, respectively. The implementation of the Classifier was performed through the “RandomForestClassifier” function in the “sklearn” Python library [61].

To optimize model performance, the hyperparameters were tuned using Grid Search with cross validation [61]. Key parameters included the 100 of trees estimators, the maximum depth of each tree 10, and the minimum number of samples required to split an internal node was set to 2. For the tuning, we used the GridSearchCV function from the “sklearn” package in Python [61], which allows for automated tuning of hyperparameters across specified parameter grids while also implementing k-fold cross-validation to ensure robust model evaluation. We set the number of folds to 5, enabling a comprehensive assessment of model performance on different subsets of the training data.

Global Sensitivity Analysis (GSA) was used to determine the influence of input variables on the output of a model [62,63]. The Sobol sensitivity analysis was performed on the RFC using the SALib Python library [64]. First, Sobol samples were generated for the input variables using the Saltelli sampling method [65]. The model was then evaluated on these samples to obtain the output probabilities and compute the first-order (S_i) and total-order (ST_i) Sobol indices. In this study, the binary output of the RFC (0 or 1) is treated as a continuous probability output, to represent probability likelihood.

3.3. Objective function

As a quality metric of the trained model we used a confusion matrix [66] and the probability of success, which is defined as the ratio between the sum of false predictions and the size of the sampled ensemble.

To ensure robust predictions, an ensemble of full physics-based models (Section 2.2) was created, with its size determined through convergence testing: Random Forest Classifier accuracy and output variance converged to a plateau beyond 1500 models, indicating sufficient parameter space coverage and justifying the choice to stop at this ensemble size.

The RFC is employed to compute the minimal distance between two wells as the minimum probability of thermal interference:

$$P = \frac{\sum_{i=1}^S (y_i)}{S} \quad (12)$$

where P is the probability of thermal interference, y_i is the outcome of a single parameter combination, and S is the sample size. To compute the probability values, we considered 100 logarithmically spaced distances between the wells from 50 to 400 m. The use of logarithmically spaced distances between 50 and 400 m is particularly beneficial when precise analysis is required for shorter distances.

4. Results

In this section we present the proof of the hypothesis of asymptotic model fit (Section 4.1), verify the quality of the surrogate model and provide the sensitivity analysis of the thermal interference (Section 4.2), and use thermal influence to estimate the minimal distance between the wells (Section 4.3), as well as hydraulic effects to estimate the maximal distance between the wells (Section 4.4).

4.1. Asymptotic fit

The asymptotic approximations (Eq. (10)–(11)) were applied to the full time series data of HRF and SE for each model in the ensemble. For visualizing the fitting results, we used the average values of HRF and SE (Fig. 5).

The mean HRF values exhibit high variability, ranging from 0.5 to 1.0 and reflecting diverse conditions modelled across the parameter space. As follows from Eq. (10) is always less or equal to 1. The HRF = 1 refers to the ideal case where all the injected thermal energy is recovered during the extraction phase of an ATEs system. For this to happen, the system has to have no heat loss to the surrounding formation, no thermal dispersion or diffusion, no ambient groundwater flow (fulfilled in this study), no buoyancy-driven heat loss, symmetrical flow paths,

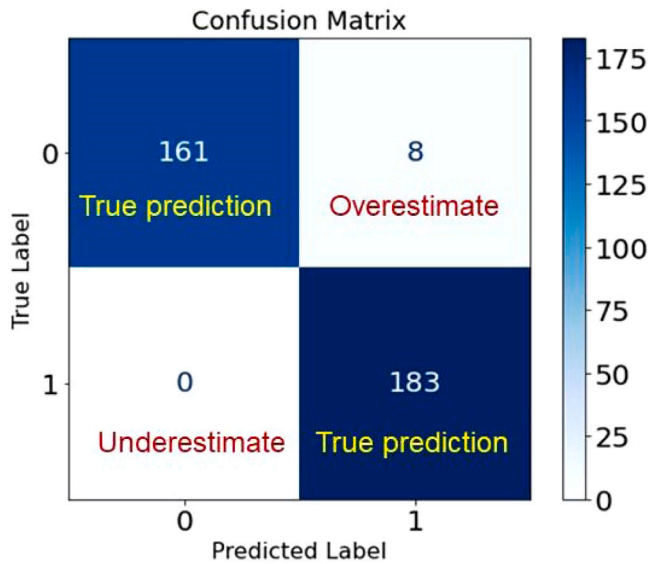


Fig. 6. Confusion matrix of the Random Forest Classifier prediction on the holdout dataset. The matrix illustrates the classification performance of RFC with the y-axis shows the true class labels derived from the physics-based model, while the x-axis indicates the predicted class labels from the RFC.

no mixing of thermal plumes with ambient groundwater. Very high HRF values (>0.8) refer to cases when the majority of these conditions have only a small influence, however, since this study considers natural conditions, therefore, HRF values are often lower. At the same time, the SE values fall within a narrower range of 0.8 to 1.0, indicating less variability in the thermal interference across the different model runs, limited by the minimum distance between wells considered in the study of 50 m. The mean HRF values computed using the asymptotic approximation closely match the results of the physics-based model, accurately representing the noted before asymptotic behaviour.

This accuracy is also observed for the SE values. The “goodness” of fit is quantified by an RMSE of 0.99 for HRF and 0.97 for SE. The parity plot (Fig. 5) further illustrates a correlation between the model outputs and their asymptotic approximations, highlighting the effectiveness of the reduced model for capturing the key performance metrics of ATEs systems.

4.2. Surrogate model and sensitivity analysis

To validate the trained RFC, 20% of the models and their associated parameter combinations were used as a holdout dataset, which did not participate in the training process. This holdout set was exclusively used to evaluate the classifier’s performance.

The potential overfitting was excluded by k-fold cross-validation and multiple random train-test splits, which ensure consistent performance and generalizability across the parameter space (Section 3.2). The validation results, illustrated in the form of a confusion matrix (Fig. 6), reflects 97% fit of the surrogate model to the physics based model. 8 out of 352 models (top-right square) tend to overestimate the thermal interference, while no model underestimates the interference. This reflects a high accuracy of the selected approximation procedure and enables the global sensitivity analysis using the surrogate model.

The Sobol sensitivity analysis [64] was performed on the RFC and technical and natural variables (Table 1). First, Sobol samples were generated for the input variables using the Saltelli sampling method [65]. The model was then evaluated on these samples to obtain the output probabilities and compute the first-order (S_i) and total-order (ST_i) Sobol indices for two scenarios: including and excluding the well distance as a variable. The results are depicted in Fig. 7.

Results of the full dataset (Fig. 7, top-left) confirm that the well distance has a first-order influence on thermal interference. The well distance has by far the highest sensitivity index of 0.9. The next influential parameters are geometry and hydraulic boundary condition of the aquifer, that is, thickness and charging flowrate.

To better resolve the effects of less dominant parameters, the same metrics computed excluding well distance from analysis, are depicted in Fig. 7, right panel. This procedure allows to avoid masking effects, as highly influential parameters can dominate the variance, making it difficult to detect the contributions of others [67]. First order sensitivity index allies with the total order sensitivity index for all parameters. When the well distance is fixed, thickness, and charging flowrate have the highest influence on the thermal interference. This makes sense since both parameters are also relevant for computing the thermal radii. The next parameters in terms of importance are the charging temperature with a total order index of about 0.3 and porosity and permeability, which share a similar result with 0.2.

The second-order sensitivity indices (Fig. 7, the bottom panels) also show the highest sensitivity for the well distance (bottom-left) along with the aquifer thickness. Besides the well distance, thickness together with permeability, as well as charging rate, are important parameters for thermal interference.

The sensitivity analysis shows that the well spacing significantly affects ATEs efficiency. This finding implies that even slight adjustments to the spacing can prevent thermal losses, directly improving energy recovery rates and reducing the overall costs associated with energy wastage and over-compensation through external energy sources.

4.3. Minimal distance between wells

Traditional well-spacing methods, such as those based on the thermal radius approach (e.g. [9,27,68]), typically rely on simplified analytical solutions to estimate the extent of thermal plumes. While these methods offer computational efficiency and intuitive design criteria, they often overlook the aquifer complexity and the impact of uncertainty in subsurface properties. In contrast, the RFC employed in this study captures nonlinear interactions between input parameters and thermal interference outcomes, offering a data-driven alternative that adapts to site-specific conditions. To compare the RFC with the thermal radius method, we ran the RFC prediction over an ensemble of possible parameters (Table 1). For each well distance, an ensemble of random parameters within the sample size of 100000 members was generated enabling convergence of a stochastic simulation. This large sample size ensures that the statistical results stabilize, and the output does not vary significantly with additional sampling. Obtained probability values at each distance are depicted in Fig. 8. The black curve represents the probability of thermal interference between wells, as derived using RFC developed in this study. The likelihood of thermal interference is the highest for distances below approximately 130 m and decreases rapidly as well spacing increases. The orange curve corresponds to corresponding thermal radius approach, enabling a direct comparison of RFC with traditional methods.

The Fig. 8 illustrates the normalized Probability Density Functions (PDFs) and histograms of different multiples of the thermal radius R_{th} . The R_{th} values were computed for each model in the ensemble of physics-based models and serve as a common measure to estimate the optimal spacing between wells in ATEs systems. The studies and regulations suggest a distance of at least $1.5 R_{th}$ between the wells to exclude thermal interference. The regulations in Netherlands suggest the well distance to be higher than $3 R_{th}$. The orange curve represents the PDF for R_{th} , while the blue and green curves show the PDFs for 1.5 and 3 times R_{th} , respectively. Corresponding histograms are depicted for each PDF, reflecting the distribution of R_{th} within the ensemble.

Based on the $1.5 R_{th}$ measure (blue curve), a well distance of more than 230 m should be generally suitable for a well placement of any ATEs doublet from the selected ensemble. If instead $3 R_{th}$

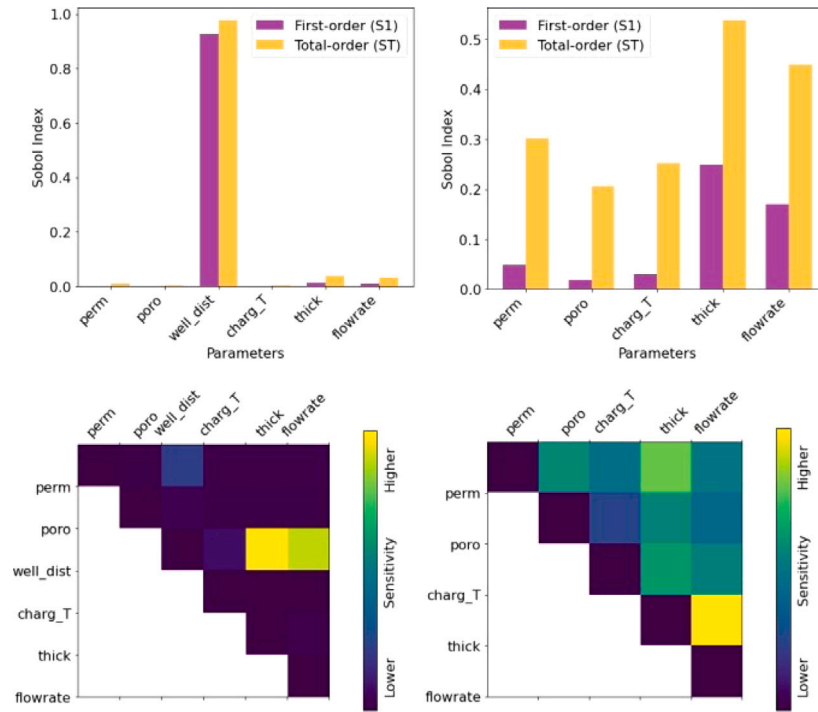


Fig. 7. Results of the global sensitivity analysis. Top: first-order Sobol indices, showing the influence of individual parameters on the output. The left panel (all parameters included) indicates the dominance of well distance, the right panel (well distance excluded) highlights the significance of charging flowrate and thickness. Bottom: heatmaps representing parameter interactions based on second-order Sobol indices, illustrating the pairwise influence of parameters on system variability. The colour intensity corresponds to the magnitude of interaction effects: lower for purple colours and higher towards yellow colours.

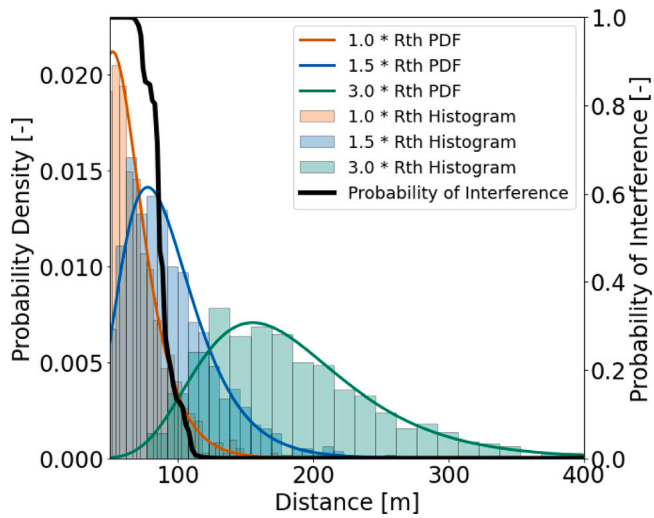


Fig. 8. Comparison of RFC-based probability of thermal interference and conventional thermal radius approach. The black line represents the probability of thermal interference estimated by the RFC, based on a stochastic analysis of physics-based model outcomes. The coloured lines (orange, blue, and green) correspond to thermal radii of 1, 1.5 and 3 R_{th} computed from the parameters ensemble. The histogram of 1.5 R_{th} refers to widely used engineering analytical approach for avoiding thermal interference.

are considered plausible, the distance between wells should generally exceed 400 m. At the same time, the results from RFC suggest that lower distances can be favourable for well placement. Within the classification task only increasing temperature in the cold well over time was considered as thermal interference while even a small increase in the cold well temperature and following decrease was classified as non-interference.

4.4. Maximal distance between the wells

Productivity Indices (PI) characterize the well performance of ATEs and are limited by the allowed pressure change in the aquifer (Eq. (8)). Closer well distance produces less pressure change and is thus favourable for hydraulic ATEs functioning. The histogram of PIs of the same ensemble of parameters as in case of RFC is depicted on Figure (9, left) with the minimum value is 0.4 m³/h/bar, the maximum 39 m³/h/bar, the mean and median are 6.8 m³/h/bar and 3.9 m³/h/bar correspondingly. With regard to the planned site in Adlershof, the preliminary estimates of a possible geological conditions suggest possible maximum drawdown of 150 m which corresponds to the minimum PI of 3.4 m³/h/bar for the flowrate of 50 m³/h, and for the minimum PI of 10 m³/h/bar for the flowrate 150 m³/h (Fig. 9, left).

The highest distance between the wells in Adlershof is limited by the maximum feasible drawdown (Eq. (8)–(9)) (which corresponds to the minimum PI) (Fig. 9). The drawdown and corresponding PI for each well distance were computed for the maximum charging flowrate 150 m³/h after 6-month cycle of ATEs for the maximum permeability of $1e^{-12}$ m² (Fig. 9, right). The drawdown increases progressively with greater well spacing due to the reduced hydraulic influence of the charging well. At the Adlershof site, this effect imposes a maximum feasible well spacing of 160 m, constrained by hydraulic and technical considerations. When combined with knowledge of thermal interference, which establishes a minimum well spacing, it is possible to define a plausible interval for well distances at the Adlershof site. This finding is important for the effective planning of ATEs systems in urban environments, as it delineates spatial constraints arising from the interplay of hydraulic and thermal properties. For the Adlershof site, the optimal well spacing lies between 130 m and 160 m. These results highlight the importance of site-specific analyses to identify feasible well placement intervals, ensuring technical performance and sustainable operation in urban conditions, providing a solid quantitative yet computationally feasible base for ATEs design and permission. Such integrated evaluations are essential for balancing efficiency, resource sustainability, and the spatial limitations often present in urban settings.

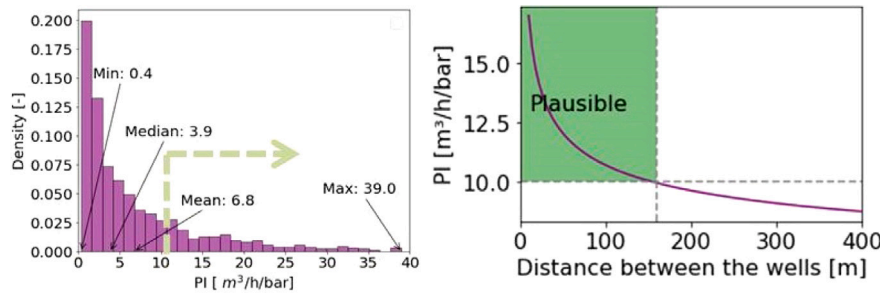


Fig. 9. Left: Histogram of Productivity Indices computed from the ensemble of physics-based models, showing the minimum (0.4 m³/h/bar), median (3.9 m³/h/bar), mean (6.8 m³/h/bar), and maximum (39.0 m³/h/bar) values. The green arrow shows PI indices plausible for the Adlershof site. Right - well distance dependent PI variation computed for the conditions of Adlershof site. The shaded green region indicates the plausible range of well distances where PI values satisfy technical and operational constraints.

5. Summary and discussion

This study presents a comprehensive approach to optimize ATEs doublet systems, integrating a novel asymptotic approximation for HRF and SE with machine learning techniques, and presenting a robust methodology for predicting and mitigating thermal interference under varying geological and technical conditions. We introduced a novel asymptotic approximation for the time development of the HRF and SE in an ATEs doublet system operating with constant six-month-long cycles. This approximation, validated across a broad range of input parameters, enables the reduction of a spatially resolved numerical model of ATEs without sacrificing accuracy. Our detailed model analysis reveals that the time derivative of Stored Energy is a critical factor in determining optimal well spacing.

A key innovation of this work is the development of an RFC trained on an ensemble of spatially resolved numerical models. The trained classifier reflects on the parameter dependencies and allows stochastic prediction of the thermal interference and global sensitivity analysis. The comparison with the conventional thermal radius approach (e.g. [9,27,68]) has demonstrated similarity with the RFC approach for a simple model (Section 4.3). While the typically applied thermal radius approach (Eq. (7)) is an effective and simple analytical estimate of the well distancing, it has several assumptions, such as simplified geology, constant boundary conditions, nonlinear effects, and long-term effects of ATEs functioning. The physics-based model used to train RFC, on the contrary, is capable of capturing these effects and complex geology, which enables a precise estimate of reliable long-term ATEs system performance. The outlined here approach provides a framework for the stochastic well distance evaluation while accounting for the model complexity while reducing computational cost by up to 10 times in comparison to Monte Carlo approaches. These results highlight the effectiveness of the proposed model in capturing the complex thermal interactions in ATEs systems, suggesting that in urbanized areas it may be profitable to implement detailed simulations in order to decide about optimal well distance. The model considered in this study serves as a demonstration of the applicability of this approach for more complex geological or technical conditions, providing a tool for the uncertainty-acknowledged planning of ATEs.

The modelling framework was applied to the planned ATEs in Berlin Adlershof, Germany. The result of probabilistic computations for an ATEs doublet shows that the probability of the thermal interference is 1 for small well distances and is constantly decreasing to 0 for the distance between the wells more than 130 m. The Productivity Index consideration, which takes into account the hydraulic response of the aquifer, suggests a maximum well distance of 160 m. Considered together, hydraulic and thermal characteristics significantly narrow down the possible range of well spacings which has to be considered in planning ATEs in urban areas. Based on such estimates, the optimal well spacing at the Adlershof site lies between 130 m and 160 m.

While the proposed model offers improvements in fast evaluation of the optimal well placement for ATEs under parametric uncertainty,

several limitations must be acknowledged to contextualize the results and inform future research.

The model assumes homogeneous aquifer properties such as permeability and porosity, which may not be representative of the highly heterogeneous geological environments where ATEs systems are implemented [12]. Although the model accounts for a broad range of parameter values, real-world aquifers may exhibit parameter variability beyond the considered ranges. This can lead to discrepancies between model predictions and actual system performance, especially in aquifers with strong stratification or fractures. Future studies should incorporate more advanced geological characterization, including field data on heterogeneity and anisotropy, to improve model accuracy.

The model neglected the influence of ambient groundwater flow, which may be non-negligible for shallow aquifers [15]. In addition, the model does not account for long-term conduction through the aquitard. By focusing primarily on advection and dispersion within the aquifer, the model may underestimate the total thermal losses, especially over extended storage periods. Including these conductive effects in future iterations of the model would provide a more comprehensive understanding of the long-term energy recovery efficiency. Even considering very slow ambient groundwater flow in the Adlershof [44,46], the next steps in site characterization therefore include groundwater isotope sampling and corresponding dating, geophysics, and tracer tests to quantify the seepage velocity.

The model operates on the assumption of consistent operational cycles, with fixed injection and extraction periods of equal duration (six months). In reality, seasonal variations in energy demand, groundwater flow, and ambient temperatures can significantly impact the performance of ATEs systems [26]. These factors may lead to suboptimal system operation if they are not considered during the design phase. A future extension of the model should explore how seasonal and climate-related variability affects ATEs efficiency, enabling more adaptive and resilient system designs.

While the model successfully optimizes technical parameters, it does not account for economic constraints or regulatory requirements that often influence the feasibility of ATEs projects [21]. Factors such as drilling costs, land ownership disputes, and regulatory compliance for well placement in urban areas can significantly alter the viability of a given design. Integrating economic models and policy frameworks into the simulation would allow for more realistic feasibility assessments.

Broader validation against a range of real-world ATEs systems is necessary to generalize the findings. Since the geological and technical settings can vary widely between sites, further field validation in diverse geological contexts is essential to confirm the model's robustness and adaptability.

While the current model provides a robust framework for optimizing ATEs system design, particularly in urban settings, future work must address these limitations to improve its real-world applicability. Incorporating more detailed geological data, accounting for long-term variability of the charging temperature and charging/extraction flowrates,

and expanding validation efforts will enhance the model's predictive power and practical relevance in varied geological contexts.

While the model's current framework is tailored to specific assumptions, future work should explore the integration of additional factors, such as seasonal temperature variations and varying storage periods, to enhance its applicability across diverse geological contexts. For the porous systems, this implies extending existing ensemble parameter variations and including in RFC additional parameters such as heterogeneity degree or statistical parameters, duration of injection/withdrawal, injection temperature, etc. The framework for fractured aquifers is generally similar, but would consider modelling within a discrete fractures network paradigm and consequently considering distributions of fracture aperture, length, and connectivity, which contributes to the system uncertainty. The next steps will focus on validation of the predictive capacity of the developed workflow by comparing with existing ATEs systems, extending the modelling approach to account for possible ambient groundwater flow and different degrees of heterogeneity, and release of the tool as an open source for urban HT-ATES planners. Calibration and validation against the field data will provide means for uncertainty reduction at later stages of ATEs planning. The present study establishes a computationally efficient and adaptable framework for improved ATEs system design, contributing to more sustainable energy systems in urban environments.

6. Conclusions

The key findings and novelty of this study can be described as follows:

- The developed analytical asymptotic approximations are capable to precisely mimic Heat Recovery Factor and Stored Energy decade-scale evolution for a 6-month injection/withdrawal cycles of ATEs over a broad parameter range.
- The developed Random Forest Classifier reflects on the parameter dependencies and allows stochastic prediction of the thermal interference and global sensitivity analysis under possible natural or technical uncertainties. In comparison to conventionally used thermal radius approach, Random Forest Classifier methodology is based on ensemble of physics-based models which can capture complex geology and dynamic conditions, providing a robust framework for reliable well distance planning in urbanized areas under parametric uncertainty.
- Until now, only the thermal consideration has been considered to identify the minimum distance. We have shown that the hydraulic consideration also helps to identify the maximum distance between the wells leading to the optimal well placement. Based on the thermal and hydraulic considerations for Berlin Adlershof, the optimal distance between the wells should be between 130 and 160 m. For optimal well design, both constraints have to be considered.
- The well spacing recommendation derived from ensemble simulations is probabilistically bounded and reflects a risk-informed decision framework based on RFC-derived probabilities.
- The global sensitivity analysis reveals that influential parameters for thermal interference are (descending order): well placement (by far), aquifer thickness, charging flowrate, charging temperature, permeability, and porosity. The model's limitations stem from the assumptions made for the spatially resolved model, the considered parameters, and their ranges. However, these constraints do not diminish the model's value in advancing ATEs design.

The study developed a generic methodology for urban planners and engineers for rapid and accurate feasibility assessments of ATEs systems under various geological and technical conditions. The Random Forest Classifier can be integrated into spatial planning tools to rapidly evaluate geothermal well-spacing scenarios under technical and natural uncertainties. The Random Forest Classifier allows for real-time screening of design alternatives, supporting zoning decisions, energy district planning, and changing strategies in urban environments. The approach enhances the efficiency and reliability of ATEs systems, particularly in densely urbanized areas where optimal well placement is crucial for minimizing energy loss and maximizing recovery efficiency.

CRediT authorship contribution statement

Elena Petrova: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Mauro Cacace:** Writing – review & editing, Software, Methodology, Investigation, Conceptualization. **Stefan Kranz:** Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Arno Zang:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Guido Blöcher:** Writing – review & editing, Visualization, Supervision, Software, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was funded the German Federal Ministry of Education and Research (BMBF) project “SpeicherCity” grant number 03G0911. We thank two anonymous reviewers for helpful comments which substantially improved the manuscript quality.

Data availability

Data will be made available on request.

References

- [1] M. Bloemendal, M. Jaxa-rozen, T. Olsthoorn, Methods for planning of ATEs systems, *Appl. Energy* 216 (December 2017) (2018) 534–557, <http://dx.doi.org/10.1016/j.apenergy.2018.02.068>.
- [2] P. Fleuchaus, B. Godschalk, I. Stober, P. Blum, Worldwide application of aquifer thermal energy storage – A review, *Renew. Sustain. Energy Rev.* 94 (2018) 861–876, <http://dx.doi.org/10.1016/j.rser.2018.06.057>.
- [3] K.S. Lee, A review on concepts, applications, and models of aquifer thermal energy storage systems, *Energies* 3 (6) (2010) 1320–1334, <http://dx.doi.org/10.3390/en3061320>.
- [4] P. Bayer, G. Attard, P. Blum, K. Menberg, The geothermal potential of cities, *Renew. Sustain. Energy Rev.* 106 (2019) 17–30, <http://dx.doi.org/10.1016/j.rser.2019.02.019>.
- [5] S. Beernink, M. Bloemendal, R. Kleinlugtenbelt, N. Hartog, Maximizing the use of aquifer thermal energy storage systems in urban areas: effects on individual system primary energy use and overall GHG emissions, *Appl. Energy* 311 (2022) 118587, <http://dx.doi.org/10.1016/j.apenergy.2022.118587>.
- [6] G. Blöcher, S. Regenspur, S. Kranz, M. Lipus, L. Pei, B. Norden, T. Reinsch, J. Hennings, R. Siemon, D. Orenczuk, S. Zeilfelder, T. Scheytt, A. Saadat, Geothermics best practices for characterization of high temperature-aquifer thermal energy storage (HT-ATES) potential using well tests in Berlin (Germany) as an example, *Geothermics* 116 (May 2022) (2024) <http://dx.doi.org/10.1016/j.geothermics.2023.102830>.
- [7] J.M. Nordbotten, Analytical solutions for aquifer thermal energy storage, *Water Resour. Res.* 53 (2) (2017) 1354–1368, <http://dx.doi.org/10.1002/2016WR019524>.
- [8] K. Chen, X. Sun, K. Soga, P.S. Nico, P.F. Dobson, L. Berkeley, Machine-learning-assisted long-term g functions for bidirectional aquifer thermal energy storage system operation, *Energy* 301 (2024) <http://dx.doi.org/10.1016/j.energy.2024.131638>.
- [9] C. Doughty, G. Hellström, C.F. Tsang, J. Claesson, A dimensionless parameter approach to the thermal behavior of an aquifer thermal energy storage system, *Water Resour. Res.* 18 (3) (1982) 571–587, <http://dx.doi.org/10.1029/WR018i003p00571>.
- [10] O. Kolditz, Modelling flow and heat transfer in fractured rocks: dimensional effect of matrix heat diffusion, *Geothermics* 24 (3) (1995) 421–437, [http://dx.doi.org/10.1016/0375-6505\(95\)00018-L](http://dx.doi.org/10.1016/0375-6505(95)00018-L).
- [11] D.W.S. Tang, S.V.D. Zee, Dispersion and recovery of solutes and heat under cyclic radial advection, *J. Hydrol.* 602 (2021) <http://dx.doi.org/10.1016/j.jhydrol.2021.126713>.

- [12] W. Sommer, J. Valstar, P. Van Gaans, T. Grotenhuis, H. Rijnaarts, The impact of aquifer heterogeneity on the performance of aquifer thermal energy storage, *Water Resour. Res.* 49 (12) (2013) 8128–8138, <http://dx.doi.org/10.1002/2013WR013677>.
- [13] Z. Song, Y. Wang, J. Wang, H. Huan, H. Li, Design of pump-and-treat strategies for contaminated groundwater remediation using numerical modeling: A case study, *Water* 16 (2024) 3665, <http://dx.doi.org/10.3390/w16243665>.
- [14] M. Antelmi, P. Mazzon, P. Hoehener, M. Marchesi, L. Alberti, Evaluation of MNA in a chlorinated solvents-contaminated aquifer using reactive transport modeling coupled with isotopic fractionation analysis, *Water* 13 (2021) 2945, <http://dx.doi.org/10.3390/w13212945>.
- [15] M. Bloemendal, T. Olsthoorn, Geothermics ATEs systems in aquifers with high ambient groundwater flow velocity, *Geothermics* 75 (January) (2018) 81–92, <http://dx.doi.org/10.1016/j.geothermics.2018.04.005>.
- [16] J.-k. Piekarski, B. Norden, S. Fuchs, S. Regensburg, L. Virchow, S. Kranz, A. Saadat, Geothermal district heating in Berlin: An ATEs field study in Berlin-Adlershof, *Proc. Eur. Geotherm. Congr. (Berl. Germany)* (October) (2022).
- [17] R. Stemmler, H. Lee, P. Blum, K. Menberg, Residential heating and cooling with aquifer thermal energy storage (ATES) on city scale, *Geotherm. Energy* 12 (March) (2023) 111560, <http://dx.doi.org/10.1186/s40517-023-00279-x>.
- [18] C.F. Tsang, T. Buscheck, C. Doughty, Aquifer thermal energy storage: A numerical simulation of Auburn University field experiments, *Water Resour. Res.* 17 (3) (1981) 647–658, <http://dx.doi.org/10.1029/WR017i003p00647>.
- [19] J. Rohmer, A. Armandine Les Landes, A. Loschetter, C. Maragna, Fast prediction of aquifer thermal energy storage: a multicyclic metamodeling procedure, *Comput. Geosci.* 27 (2) (2023) 223–243, <http://dx.doi.org/10.1007/s10596-023-10192-8>.
- [20] M. Ueckert, T. Baumann, Hydrochemical aspects of high-temperature aquifer storage in carbonaceous aquifers: evaluation of a field study, *Geotherm. Energy* 7 (1) (2019) <http://dx.doi.org/10.1186/s40517-019-0120-0>.
- [21] S. Schüppler, P. Fleuchaus, P. Blum, Techno-economic and environmental analysis of an aquifer thermal energy storage (ATES) in Germany, *Geotherm. Energy* (2019) <http://dx.doi.org/10.1186/s40517-019-0127-6>.
- [22] L. Gao, J. Zhao, Q. An, J. Wang, X. Liu, A review on system performance studies of aquifer thermal energy storage, *Energy Procedia* 142 (2017) 3537–3545, <http://dx.doi.org/10.1016/j.egypro.2017.12.242>.
- [23] A. Yapparova, S. Matth, Realistic simulation of an aquifer thermal energy storage: Effects of injection temperature, well placement and groundwater flow, *Energy* 76 (2014) 1011–1018, <http://dx.doi.org/10.1016/j.energy.2014.09.018>.
- [24] R. Duijff, M. Bloemendal, M. Bakker, Interaction effects between aquifer thermal energy storage systems, *Groundwater* 61 (2) (2021) 173–182, <http://dx.doi.org/10.1111/gwat.13163>.
- [25] P. Blum, G. Campillo, T. Kolbel, Techno-economic and spatial analysis of vertical ground source heat pump systems in Germany, *Energy* 36 (5) (2011) 3002–3011, <http://dx.doi.org/10.1016/j.energy.2011.02.044>.
- [26] W. Sommer, J. Valstar, I. Leusbrock, T. Grotenhuis, H. Rijnaarts, Optimization and spatial pattern of large-scale aquifer thermal energy storage, *Appl. Energy* 137 (2015) 322–337, <http://dx.doi.org/10.1016/j.apenergy.2014.10.019>.
- [27] S. Beernink, N. Hartog, P.J. Vardon, M. Bloemendal, Geothermics heat losses in ATEs systems: The impact of processes, storage geometry and temperature, *Geothermics* 117 (November 2023) (2024) 102889, <http://dx.doi.org/10.1016/j.geothermics.2023.102889>.
- [28] M. Chen, A.F.B. Tompson, R.J. Mellors, O. Abdalla, An efficient optimization of well placement and control for a geothermal prospect under geological uncertainty, *Appl. Energy* 137 (2015) 352–363, <http://dx.doi.org/10.1016/j.apenergy.2014.10.036>.
- [29] T. Hermans, F. Nguyen, M. Klepikova, A. Dassargues, J. Caers, Uncertainty quantification of medium-term heat storage from short-term geophysical experiments using Bayesian evidential learning, *Water Resour. Res.* 54 (4) (2018) 2931–2948, <http://dx.doi.org/10.1002/2017WR022135>.
- [30] P. Selzer, H. Shao, C. Behrens, C. Lehmann, R. Seydewitz, R. Lu, P. Kreye, W. Rühaak, O. Kolditz, The value of simplified models of radionuclide transport for the safety assessment of nuclear waste repositories: A benchmark study, *J. Contam. Hydrol.* 267 (2024) 104417, <http://dx.doi.org/10.1016/j.jconhyd.2024.104417>.
- [31] I. Gomes Goncalves, F. Wellmann, Uncertainty propagation in deep Gaussian process networks, *Math. Geosci.* (2025) <http://dx.doi.org/10.1007/s11004-025-10187-4>.
- [32] R. Stemmler, A. Arab, S. Bauer, C. Beyer, G. Blöcher, C. Bossennec, M. Dörnbrack, F. Hahn, P. Jaeger, S. Kranz, A. Mauerberger, J.N. Nordheim, M. Ohagen, E. Petrova, S. Regensburg, D. Rettenmaier, A. Saadat, I. Sass, T. Scheytt, N. Scholliers, H. Shao, K. Tzoufka, K. Zosseder, P. Blum, Current research on aquifer thermal energy storage (ATES) in Germany, *Grundwasser* (2025) <http://dx.doi.org/10.1007/s00767-025-00590-3>.
- [33] M. Abuasbeh, J. Acuna, A. Lazzarotto, B. Palm, Long term performance monitoring and KPIs' evaluation of aquifer thermal energy storage system in esker formation: Case study in Stockholm, *Geothermics* 96 (2021) 102166, <http://dx.doi.org/10.1016/j.geothermics.2021.102166>.
- [34] H.B. Pratama, M.C. Supijo, Sutopo, Experimental design and response surface method in geothermal energy: A comprehensive study in probabilistic resource assessment, *Geothermics* 87 (2020) 101869, <http://dx.doi.org/10.1016/j.geothermics.2020.101869>.
- [35] J. Allgeier, O.A. Cirpka, Surrogate-model assisted plausibility-check, calibration, and posterior-distribution evaluation of subsurface-flow models, *Water Resour. Res.* 59 (7) (2023) <http://dx.doi.org/10.1029/2023WR034453>.
- [36] R.G. Regis, C.A. Shoemaker, A stochastic radial basis function method for the global optimization of expensive functions, *INFORMS J. Comput.* 19 (4) (2007) 497–509, <http://dx.doi.org/10.1287/ijoc.1060.0182>.
- [37] G.G. Wang, S. Shan, Review of metamodeling techniques in support of engineering design optimization, *J. Mech. Des.* 129 (4) (2006) 370–380, <http://dx.doi.org/10.1115/1.2429697>.
- [38] J. Manceau, J. Rohmer, Post-injection trapping of mobile CO₂ in deep aquifers: Assessing the importance of model and parameter, *Comput. Geosci.* (2016) 1251–1267, <http://dx.doi.org/10.1007/s10596-016-9588-x>.
- [39] Y. Wang, F. Zhang, Operation optimization for aquifer thermal energy storage (ATES) systems based on a surrogate model - assisted method, *Appl. Therm. Eng.* 268 (2025) 125907, <http://dx.doi.org/10.1016/j.applthermaleng.2025.125907>.
- [40] L. Li, T. Zhou, J. Li, X. Wang, A machine learning-based decision support framework for energy storage selection, *Chem. Eng. Res. Des.* 181 (2022) 412–422, <http://dx.doi.org/10.1016/j.cherd.2022.04.023>.
- [41] L. Breiman, Random forests, *Mach. Learn.* 45 (45) (2001) 5–32, <http://dx.doi.org/10.1023/A:1010933404324>.
- [42] M. Frick, M. Scheck-Wenderoth, M. Schneider, M. Cacace, Surface to groundwater interactions beneath the City of Berlin: Results from 3D models, *Geofluids* 2019 (1) (2019) 4129016, <http://dx.doi.org/10.1155/2019/4129016>.
- [43] L. Pei, L. Virchow, G. Blöcher, S. Kranz, A. Saadat, Characterization of artesian flow and heat transition in an ATEs research wellbore using DTS monitoring and numerical modelling, *Adv. Geosci.* 65 (2024) 37–44, <http://dx.doi.org/10.5194/adgeo-65-37-2024>.
- [44] K.H. Hebig, N. Ito, T. Scheytt, A. Marui, Review: Deep groundwater research with focus on Germany, *Hydrogeol. J.* (2005) <http://dx.doi.org/10.1007/s10040-011-0815-1>.
- [45] M. Frick, M. Scheck-Wenderoth, M. Cacace, M. Schneider, 3D thermohydraulic model for Berlin's Urban Subsurface, 2024, <http://dx.doi.org/10.5281/zenodo.14002255>, Data set.
- [46] M. Frick, M. Scheck-Wenderoth, M. Cacace, M. Schneider, Boundary condition control on inter-aquifer flow in the subsurface of Berlin (Germany) – new insights from 3-D numerical modelling, *Adv. Geosci.* 49 (2019) 9–18, <http://dx.doi.org/10.5194/adgeo-49-9-2019>.
- [47] C. Geuzaine, J.-F. Remacle, Gmsh: a three-dimensional finite element mesh generator with built-in pre- and post-processing facilities, *Internat. J. Numer. Methods Engrg.* (2009) 1–24, <http://dx.doi.org/10.1002/nme.2579>.
- [48] G. Giudicelli, A. Lindsay, L. Harbour, C. Icenhour, M. Li, J.E. Hansel, P. German, P. Behne, O. Marin, R.H. Stogner, J.M. Miller, D. Schwen, Y. Wang, L. Munday, S. Schunert, B.W. Spencer, D. Yushu, A. Recuero, Z.M. Prince, M. Nezdur, T. Hu, Y. Miao, Y.S. Jung, C. Matthews, A. Novak, B. Langley, T. Truster, N. Nobre, B. Alger, D. Andrić, F. Kong, R. Carlsen, A.E. Slaughter, J.W. Peterson, D. Gaston, C. Permann, 3.0 - MOOSE: Enabling massively parallel multiphysics simulations, *SoftwareX* (2024).
- [49] M. Cacace, A.B. Jacquey, Flexible parallel implicit modelling of coupled thermal-hydraulic-mechanical processes in fractured rocks, *Solid Earth* 8 (5) (2017) 921–941, <http://dx.doi.org/10.5194/se-8-921-2017>.
- [50] IAPWS, Release on the IAPWS formulation 2008 for the viscosity of ordinary water substance, *Tech. Rep.* (2008) 2931–2948, doi:<http://www.iapws.org.10>.
- [51] M. Bakr, N. van Oostrom, W. Sommer, Efficiency of and interference among multiple aquifer thermal energy storage systems; A Dutch case study, *Renew. Energy* 60 (2013) 53–62, <http://dx.doi.org/10.1016/j.renene.2013.04.004>.
- [52] S. Kranz, J. Bartels, Simulation and data based optimisation of an operating seasonal aquifer thermal energy storage, *World Geotherm. Congr. Bali Indones. (April)* (2010) 25–29.
- [53] C. Wang, X. Qiang, M. Xu, T. Wu, Recent advances in surrogate modeling methods for uncertainty quantification and propagation, *Symmetry* 14 (6) (2022) <http://dx.doi.org/10.3390/sym14061219>.
- [54] C. Hou, K. Behdinan, Dimensionality reduction in surrogate modeling: A review of combined methods, *Data Sci. Eng.* 7 (2022) <http://dx.doi.org/10.1007/s41019-022-00193-5>.
- [55] M. Vohra, P. Nath, S. Mahadevan, Y.-t.T. Lee, Fast surrogate modeling using dimensionality reduction in model inputs and field output: Application to additive manufacturing, *Reliab. Eng. Syst. Saf.* 201 (May) (2020) <http://dx.doi.org/10.1016/j.ress.2020.106986>.
- [56] P. Virtanen, R. Gommers, T.E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S.J. van der Walt, M. Brett, J. Wilson, K.J. Millman, N. Mayorov, A.R.J. Nelson, E. Jones, R. Kern, E. Larson, C. Carey, I. Polat, Y. Feng, E.W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E.A. Quintero, C.R. Harris, A.M. Archibald, A.H. Ribeiro, F. Pedregosa, P. van Mulbregt, Y. Vázquez-Baeza, SciPy 1.0: Fundamental algorithms for scientific computing in python, *Nature Methods* 17 (2020) 261–272, <http://dx.doi.org/10.1038/s41592-019-0686-2>.

- [57] S.B. Nadkarni, G.S. Vijay, R.C. Kamath, Comparative study of random forest and gradient boosting algorithms to predict airfoil self-noise, *Eng. Proc.* 59 (1) (2023) <http://dx.doi.org/10.3390/engproc2023059024>.
- [58] J. Wainer, Comparison of 14 different families of classification algorithms on 115 binary datasets, 2016, <http://dx.doi.org/10.48550/arXiv.1606.00930>, arXiv.
- [59] T. Adugna, W. Xu, J. Fan, Comparison of random forest and support vector machine classifiers for regional land cover mapping using coarse resolution FY-3C images, *Remote. Sens.* 14 (3) (2022) <http://dx.doi.org/10.3390/rs14030574>.
- [60] H.L. Smith, P.J. Biggs, N.P. French, A.N.H. Smith, J.C. Marshall, Out of (the) bag—encoding categorical predictors impacts out-of-bag samples, *PeerJ Comput. Sci.* (10:e2445) (2024) <http://dx.doi.org/10.7717/peerj-cs.2445>.
- [61] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, É. Duchesnay, Scikit-learn: Machine learning in python, *J. Mach. Learn. Res.* 12 (2011) 2825–2830.
- [62] I.M. Sobol, A Primer for the Carlo Method, CRC Press, 1994, p. 107, <http://dx.doi.org/10.1201/9781315136448>.
- [63] G.E. Archer, A. Saltelli, I.M. Sobol, Sensitivity measures, anova-like techniques and the use of bootstrap, *J. Stat. Comput. Simul.* 58 (2) (1997) 99–120, <http://dx.doi.org/10.1080/00949659708811825>.
- [64] J. Herman, W. Usher, SALib: An open-source python library for sensitivity analysis, *J. Open Source Softw.* 2 (9) (2017) 97, <http://dx.doi.org/10.21105/joss.00097>.
- [65] A. Saltelli, Making best use of model evaluations to compute sensitivity indices, *Comput. Phys. Comm.* 145 (2) (2002) 280–297.
- [66] S.V. Stehman, Selecting and interpreting measures of thematic classification accuracy, *Remote Sens. Environ.* 62 (1) (1997) [http://dx.doi.org/10.1016/S0034-4257\(97\)00083-7](http://dx.doi.org/10.1016/S0034-4257(97)00083-7).
- [67] A. Saltelli, M. Ratto, T. Andres, F. Campolongo, J. Cariboni, D. Gatelli, M. Saisana, S. Tarantola, Global Sensitivity Analysis: The Primer, John Wiley & Sons, 2008, <http://dx.doi.org/10.1002/9780470725184>.
- [68] H.A. Sheldon, A. Wilkins, C.P. Green, Recovery efficiency in high-temperature aquifer thermal energy storage systems, *Geothermics* 96 (March) (2021) <http://dx.doi.org/10.1016/j.geothermics.2021.102173>.